Year 12 Further Maths - Power and resistance

Similar questions - Vehicles, riders and boats

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 A lorry of mass 6000 kg has an engine working at a constant rate of 90 kW. When the lorry moves on a straight horizontal road it experiences a constant resistance of 1800 N.

(a) Find the maximum speed of the lorry. **[2]**

(b) Find the acceleration of the lorry when its speed is 20 $\mathrm{m\,s^{-1}}$. **[3]**

The lorry now pulls a trailer of mass 2000 kg along the same road. The resistance to the motion of the trailer is a constant 600 N, and the resistance on the lorry is unchanged.

(c) Find the new maximum speed of the lorry and trailer. **[2]**

(d) Find the tension in the coupling when the speed is 30 $\mathrm{m\,s^{-1}}$. **[4]**

(e) At this instant the coupling breaks. Assuming the resistance on the trailer remains 600 N, find how far the trailer travels before coming to rest. **[3]**

2 A cyclist and her bicycle have a combined mass of 80 kg. The resistance to motion is modelled as kv N, where $v\,\mathrm{m\,s^{-1}}$ is the speed. On a straight horizontal road, working at a constant rate of 360 W, she has a maximum speed of 12 $\mathrm{m\,s^{-1}}$.

(a) Show that $k = 2.5$. **[2]**

(b) She now rides up a straight hill inclined at $\sin^{-1} 0.05$ to the horizontal, again working at 360 W. Find her maximum speed up the hill. **[5]**

(c) Find her acceleration up the hill when her speed is 4 $\mathrm{m\,s^{-1}}$. **[3]**

(d) At the top she stops pedalling and freewheels down the same hill. Find the speed she approaches, according to the model. **[3]**

3 A kayaker and her kayak have a combined mass of 200 kg. The maximum power she can generate is 100 W. In a model of the motion the total resistance is a constant 80 N whenever the kayak is moving.

(a) Find the maximum possible acceleration of the kayak, according to the model, when its speed is 0.4 $\mathrm{m\,s^{-1}}$. **[3]**

(b) At one stage the kayak travels at constant speed with the kayaker generating 120 W (this is assumed constant). The kayak passes a buoy and then a second buoy 280 m further on. Find how long, according to the model, the kayak takes to travel between the buoys. **[4]**

(c) Give one reason why assuming that the kayaker's power is constant may be unrealistic. **[1]**

(d) On another occasion the kayak moves in a straight line while she generates a constant 112 W. Its speed increases from 1.0 $\mathrm{m\,s^{-1}}$ to 1.2 $\mathrm{m\,s^{-1}}$ over a distance of 90 m. Use an energy method to find the time taken. **[5]**

4 A scooter and rider have total mass 150 kg. The resistance to motion is proportional to the speed. With the engine working at a constant 3 kW, the maximum speed on a straight horizontal road is 20 $\mathrm{m\,s^{-1}}$.

(a) Find the speed of the scooter when it is working at 3 kW and its acceleration is 0.8 $\mathrm{m\,s^{-2}}$. **[7]**

(b) Find the maximum speed, working at 3 kW, up a straight road inclined at $\sin^{-1} 0.04$ to the horizontal. **[5]**

Question 1

(a) $v = P/R = 90000/1800 = 50\ \text{m s}^{-1}$

(b) Driving force $= 90000/20 = 4500$ N; $6000a = 4500 - 1800$, so $a = 0.45\ \text{m s}^{-2}$

(c) Total resistance 2400 N: $v = 90000/2400 = 37.5\ \text{m s}^{-1}$

(d) Driving force $= 90000/30 = 3000$ N. Whole system: $8000a = 3000 - 2400$, $a = 0.075$. Trailer: $T - 600 = 2000(0.075)$, so $T = 750$ N

(e) Deceleration of trailer $= 600/2000 = 0.3\ \text{m s}^{-2}$; $0 = 30^2 - 2(0.3)s$, so $s = 1500$ m

Question 2

(a) At maximum speed driving force equals resistance: $360/12 = 12k$, so $k = 2.5$

(b) At maximum speed up the hill: $\dfrac{360}{v} = 2.5v + 80g(0.05) = 2.5v + 39.2$. So $2.5v^2 + 39.2v - 360 = 0$ and $v = 6.49\ \text{m s}^{-1}$ (3 s.f.)

(c) $\dfrac{360}{4} - 2.5(4) - 39.2 = 80a$: $a = 0.51\ \text{m s}^{-2}$

(d) Terminal speed when $2.5v = 39.2$: $v = 15.7\ \text{m s}^{-1}$ (15.68)

Question 3

(a) Driving force $= 100/0.4 = 250$ N. $200a = 250 - 80$, so $a = 0.85\ \text{m s}^{-2}$

(b) Constant speed: $120/v = 80$, so $v = 1.5\ \text{m s}^{-1}$. Time $= 280/1.5 = 187$ s (186.7)

(c) For example: a person cannot maintain a constant rate of working (tiring, stroke cycle), or the power is not constant when she accelerates

(d) Work–energy: $112t - 80(90) = \frac{1}{2}(200)(1.2^2 - 1.0^2) = 44$. So $112t = 7244$ and $t = 64.7$ s (64.68)

Question 4

(a) $R = kv$ and at $20\ \text{m s}^{-1}$: $3000 = 20(20k)$, so $k = 7.5$. Then $\dfrac{3000}{v} - 7.5v = 150(0.8) = 120$, giving $7.5v^2 + 120v - 3000 = 0$, i.e. $v^2 + 16v - 400 = 0$ and $v = 13.5\ \text{m s}^{-1}$ (13.54)

(b) $150g(0.04) = 58.8$ N. $\dfrac{3000}{v} = 7.5v + 58.8$, so $7.5v^2 + 58.8v - 3000 = 0$ and $v = 16.5\ \text{m s}^{-1}$ (16.46)